

Compatibility Validation Module (CVM)

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Canonical Definition

Compatibility Validation Module (CVM) defines the structural conditions under which autonomous operational compatibility is verified, validated, monitored, and continuously governed before and during interoperability between autonomous operational systems.

CVM governs compatibility validation.

The module establishes the governance conditions required to ensure that autonomous operational actors satisfy compatibility requirements before trusted operational cooperation begins and throughout continued operation.

Compatibility should not be assumed.

Compatibility should be validated.

CVM governs that validation.

Module Operational Role

CVM defines the compatibility validation layer of ACPS.

Its role is to verify that autonomous operational systems remain operationally and governance compatible before and during autonomous

cooperation.

Module Operational Space

CVM governs:

- compatibility validation,
 - interoperability validation,
 - governance compatibility verification,
 - operational compatibility assessment,
 - compatibility monitoring,
 - autonomous ecosystem validation,
 - governed compatibility assurance,
 - continuous compatibility verification.
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Module Function

The module applies wherever organizations must validate compatibility before autonomous operational systems exchange information, coordinate activities, or perform joint operations.

Minimum Implementation Framework

1. Define the Compatibility Validation Object Identify which autonomous operational relationships require compatibility validation.

2. Define Validation Conditions

Establish governance conditions governing compatibility verification before and during autonomous cooperation.

3. Define Validation Failure Logic

Identify conditions where compatibility becomes unverified, degraded, inconsistent, incomplete, or governance-invalid.

4. Define Governance Response

Define governance actions for compatibility approval, operational restriction, corrective adaptation, revalidation, escalation, or suspension of autonomous cooperation.

5. Preserve Validation Traceability

Preserve compatibility assessments, governance decisions, operational evidence, validation history, monitoring records, and supporting documentation.

Use Case 1 — Autonomous Airport Operations

Scenario

Autonomous baggage systems, security robots, and aircraft ground-support vehicles coordinate operations across a major international airport.

Application

CVM validates operational and governance compatibility before autonomous collaboration is authorized.

Result

Airport operations remain interoperable, reliable, and continuously governed.

Use Case 2 — Multi-Vendor Autonomous Mining Operation

Scenario

Autonomous drilling systems, transport vehicles, and safety robots supplied by different manufacturers collaborate in a mining environment.

Application

CVM continuously validates compatibility as operational conditions and system configurations evolve.

Result

The mining operation maintains safe, trustworthy, and governance-approved interoperability across the complete autonomous ecosystem.

Canonical Closing Statement

Governable compatibility requires continuous validation.

Compatibility Validation ensures that autonomous operational systems remain interoperable, governance-compatible, and operationally trustworthy before and throughout autonomous cooperation.

Related Documents

→ [Autonomous Compatibility Standard \(ACPS\)](#)

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Canonical Source

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